

Hypothesis Testing for One Sample Proportion

Statistics Worksheet · Grade 11–12 / Intro College Stats

Name: _____

Date: _____

Learning Objectives

- Write null and alternative hypotheses for a one-sample proportion test
- Verify conditions (normality, independence, randomness) before conducting a test
- Calculate the z-test statistic and use the p-value to draw a conclusion

Problems

1. A company claims that 60% of its customers are satisfied with their service. Identify the null and alternative hypotheses if a researcher wants to test whether the true proportion differs from 60%.

$$H_0 : p = 0.60 \quad H_a : p \neq 0.60$$

2. A politician claims that more than 50% of voters support her. Write the null and alternative hypotheses for this claim.

$$H_0 : p = 0.50 \quad H_a : p > 0.50$$

3. A sample of 80 students is drawn. The hypothesized population proportion is $p = 0.40$. Check whether the normality condition is satisfied using the rule: $n \cdot p > 10$ and $n \cdot q > 10$.

$$n = 80, \quad p = 0.40, \quad q = 1 - 0.40$$

4. A survey of 50 people finds that 18 prefer brand A. Calculate the sample proportion ($\hat{p}$).

$$\hat{p} = \frac{x}{n} = \frac{18}{50}$$

5. Using a sample of $n = 100$ with $\hat{p} = 0.55$ and a hypothesized population proportion of $p = 0.50$, calculate the z-test statistic.

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$



6. Ryan, a basketball player, made 40% of his free throws last season. This season he made 25 out of 40 attempts. Test at the 1% significance level whether his proportion has improved. First, write the hypotheses and check the normality condition.

$$H_0 : p = 0.40 \quad H_a : p > 0.40$$

7. Continuing the Ryan example: $p = 0.40$, $n = 40$, and he made 25 out of 40 free throws. Calculate the z-test statistic.

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} = \frac{\frac{25}{40} - 0.40}{\sqrt{\frac{(0.40)(0.60)}{40}}}$$

8. Using the z-test statistic of approximately 3.23 from the Ryan free throw problem and a right-tailed test at the 1% significance level ($\alpha = 0.01$, critical $z = 2.326$), state the conclusion. The p-value for $z = 3.23$ is approximately 0.0006.

$$\alpha = 0.01, \quad z = 3.23, \quad p\text{-value} \approx 0.0006$$

9. A health agency claims that less than 25% of adults in a city exercise regularly. A random sample of 200 adults finds that 44 exercise regularly. At the 5% significance level, test the agency's claim. Show all four steps: (1) hypotheses, (2) conditions, (3) test statistic, (4) conclusion. The p-value for this test is approximately 0.1335.

$$H_0 : p = 0.25 \quad H_a : p < 0.25$$

10. A manufacturer claims that exactly 90% of its light bulbs last more than 1,000 hours. A quality inspector randomly tests 150 bulbs and finds that 126 last more than 1,000 hours. At the 5% significance level, test whether the true proportion differs from 90%. The p-value for this test is approximately 0.0241. State all four steps, identify the type of test, and write a full conclusion in context.

$$H_0 : p = 0.90 \quad H_a : p \neq 0.90$$



Hypothesis Testing for One Sample Proportion — Answer Key

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Answer Key

1. Answer: $H_0: p = 0.60$ (two-tailed test)

- The null hypothesis always uses an equals sign: $H_0: p = 0.60$
- Because the researcher tests whether the proportion 'differs' (not simply higher or lower), the alternative is two-tailed: $H_a: p \neq 0.60$

2. Answer: $H_0: p = 0.50$, $H_a: p > 0.50$ (right-tailed test)

- The null hypothesis states the proportion equals the claimed value: $H_0: p = 0.50$
- Because the claim is 'more than,' the alternative hypothesis is right-tailed: $H_a: p > 0.50$

3. Answer: Normality condition is satisfied: $np = 32 > 10$ and $nq = 48 > 10$

- Compute $q = 1 - 0.40 = 0.60$
- $n \cdot p = 80 \times 0.40 = 32 > 10$ ✓
- $n \cdot q = 80 \times 0.60 = 48 > 10$ ✓
- Both conditions are met, so normality is satisfied.

4. Answer: $\hat{p} = 0.36$

- Divide the number of successes by the sample size: $\hat{p} = 18 \div 50$
- $\hat{p} = 0.36$

5. Answer: $z \approx 1.00$

- Identify values: $\hat{p} = 0.55$, $p = 0.50$, $q = 0.50$, $n = 100$
- Compute standard deviation: $\sqrt{(0.50 \times 0.50 / 100)} = \sqrt{(0.0025)} = 0.05$
- $z = (0.55 - 0.50) / 0.05 = 0.05 / 0.05 = 1.00$

6. Answer: $H_0: p = 0.40$ (right-tailed); $np = 16 > 10$, $nq = 24 > 10$ — normality satisfied

- The parameter p = proportion of Ryan's successful free throws
- $H_0: p = 0.40$ (no improvement); $H_a: p > 0.40$ (improvement) — right-tailed
- Check normality: $n \cdot p = 40 \times 0.40 = 16 > 10$ ✓ and $n \cdot q = 40 \times 0.60 = 24 > 10$ ✓
- Normality condition is satisfied.

7. Answer: $z \approx 3.23$

- $\hat{p} = 25/40 = 0.625$
- Standard deviation = $\sqrt{(0.40 \times 0.60 / 40)} = \sqrt{(0.24/40)} = \sqrt{0.006} \approx 0.07746$
- $z = (0.625 - 0.40) / 0.07746 = 0.225 / 0.07746 \approx 3.23$

8. Answer: Reject H_0 . There is sufficient evidence that Ryan's free throw proportion has improved.

- Compare the p-value to α : $0.0006 < 0.01$



- Since $p\text{-value} < \alpha$, we reject the null hypothesis.
- Conclusion: At the 1% significance level, there is sufficient evidence to conclude that Ryan's free throw proportion has increased above 40%.

9. Answer: Fail to reject H_0 ; insufficient evidence that fewer than 25% exercise regularly

- Step 1 — Hypotheses: $H_0: p = 0.25$, $H_a: p < 0.25$ (left-tailed); p = proportion of adults who exercise regularly
- Step 2 — Conditions: $n \cdot p = 200 \times 0.25 = 50 > 10$ ✓; $n \cdot q = 200 \times 0.75 = 150 > 10$ ✓; sample is random ✓
- Step 3 — Test statistic: $p = 44/200 = 0.22$; $SE = \sqrt{(0.25 \times 0.75/200)} = \sqrt{0.0009375} \approx 0.03062$; $z = (0.22 - 0.25)/0.03062 \approx -0.98$
- Step 4 — Conclusion: $p\text{-value} \approx 0.1335 > 0.05 = \alpha$; fail to reject H_0 . There is not sufficient evidence to support the claim.

10. Answer: Reject H_0 ; sufficient evidence that the true proportion differs from 90%

- Step 1 — Hypotheses: $H_0: p = 0.90$, $H_a: p \neq 0.90$ (two-tailed); p = proportion of bulbs lasting more than 1,000 hours
 - Step 2 — Conditions: $n \cdot p = 150 \times 0.90 = 135 > 10$ ✓; $n \cdot q = 150 \times 0.10 = 15 > 10$ ✓; random sample ✓
 - Step 3 — Test statistic: $p = 126/150 = 0.84$; $SE = \sqrt{(0.90 \times 0.10/150)} = \sqrt{0.0006} \approx 0.02449$; $z = (0.84 - 0.90)/0.02449 \approx -2.45$
 - Step 4 — Conclusion: $p\text{-value} \approx 0.0241 < 0.05 = \alpha$; reject H_0 . At the 5% significance level, there is sufficient evidence that the true proportion of bulbs lasting more than 1,000 hours differs from the manufacturer's claimed 90%.
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